

A Smart System for Early Detection and Prevention of Forest Fire using UAV Swarms and VLM

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Abstract—The need for sustainability is on the rise as forests are critical to ecosystem stabilization, biodiversity, and climate change protection. Forest fires are a threat to these ecosystems, particularly in the hot climate areas. Traditional monitoring tools, such as the Internet of Things (IoT) systems, are expensive and hard to implement in nature-dependent landscapes or in remote areas. In this paper, we introduce a smart and scalable system for early detection and prevention of forest fires with a swarm of UAVs, based on a spiral coverage algorithm to monitor large forest areas and adapt to environmental conditions such as high temperatures. Vision-Language Models (VLMs) are used to analyze images and detect the early signs of fire (e.g., smoke, flames, or abnormal thermal changes). If any of them is detected, an alarm is triggered, and a preventive action is taken with UAVs equipped with fire-extinguishing agents to stop the spread of the fire. This approach is cost-effective and flexible, leading to improved early warning and forest conservation.

Keywords—Smart Monitoring, Drone, Sustainability, AI, Urban Green Spaces, Fog Computing

I. INTRODUCTION

Sustainability is a fundamental part of global environmental and developmental planning, research, and policymaking, and it is increasingly emphasized in our global efforts to protect our natural resources and reduce the impact of climate change on the human environment [1, 2]. Nations are working to integrate sustainability in areas like smart transport [3], health care [4, 5], and natural resources.

Natural ecosystems like forests are at the heart of sustainability because they are crucial to maintain ecological balance and biodiversity and to sustain life on Earth. Forests are carbon sinks and contribute to oxygen production and climate control, as well as food, timber, and medicinal crops,

while sustaining biodiversity [6]. However, forests are also in a state of crisis worldwide (forest fires are among the most serious, life-threatening, and rapidly spreading environmental disasters).

The intensity and frequency of these fires increased significantly in recent years because of climate change, rising temperatures, long droughts, and the difficulty of early identification of ignition zones [7, 8]. Some of these fires have destroyed vast areas of forest in many countries worldwide, such as Australia, the United States, and parts of the Middle East, causing both environmental and economic destruction, biodiversity loss, and damage to ecosystems [9].

Today, forest fire monitoring systems are largely based on satellite images, ground-based surveillance systems, IoT, and manual observation by forest rangers and human patrols. They are used in many aspects to control forest fires and are useful; however, there are some difficulties in using them - the time taken to detect fires, the high cost of installation and maintenance of sensors in large areas, and the difficulty of implementation in dense or remote areas. In addition, many of them are not suitable to respond to crises in real emergencies in a timely manner [12].

In this regard, UAVs have emerged as a promising solution in environmental monitoring because of their strong mobility, rugged terrain coverage, and their ability to cover extensive areas efficiently [13]. A swarm of UAVs can provide better and continuous coverage than a single drone. Moreover, AI has advanced the capabilities of these aerial vehicles to analyze the data in real time [14, 15].

Vision-language models (VLMs) are one of the most advanced technologies capable of understanding and analyzing visual scenes. These architectures are pre-trained on massive

datasets and are very useful to detect early indicators of fires, such as smoke, flames, or any other abnormal changes in the forest. The combination of VLMs and UAVs can enable smart, adaptive, and effective monitoring with fast response for any case [16].

Saudi Arabia considers environmental sustainability and saving natural resources to be the main goals of development to create a sustainable life and the well-being of its citizens. There are many national initiatives in this context, and one of the most important initiatives is the Saudi Green, which targets reducing carbon, increasing dependence on clean energy, and increasing the vegetation ratio in different areas. Another initiative is planting billions of trees and rehabilitating degraded land to face desertification, support biodiversity, and enhance air quality [17]. Moreover, Saudi Arabia depends on developing advanced methodologies to save and secure forests and vegetation from environmental threats like fires or floods, by adopting modern technologies and finding early monitoring and response models.

Although the important efforts of Saudi Antitaves and contributions reflect a high recognition of the importance of forests and vegetation, which is not only for environmental resources, but also for long-term sustainability. However, there is still a need for more smart solutions and modern technologies, such as UAVs, Digital Twin, AI, etc. (to name a few). These technologies are required to support any environmental conservation process and to keep the green agenda going [18].

In this research, we propose a new approach to early detection and prevention of forest fires using a swarm of UAVs based on a spiral coverage algorithm, which ensures continuous and regular monitoring of forested areas. The system can also adjust to the environment by increasing the frequency of monitoring when temperatures are high. Also, VLMs are used to analyze the images and detect early fire indicators, and quickly apply preventive measures with drones equipped with fire-extinguishing materials to prevent fire propagation.

This approach aims to combat the drawbacks of current systems and provide a flexible, scalable, and cost-effective solution that combines drone swarm smart with AI technology for forest preservation and a sustainable environment in the long run.

Consequently, the main contributions of this research are as follows:

- Propose a smart framework for the early detection of forest fires using a swarm of UAVs.
- Develop a Spiral Coverage algorithm to ensure the efficient monitoring of vast forested areas.
- Adopt VLMs for fast image analysis tasks to detect any early fire indicators.
- Propose a response mechanism via drones to proactively deploy fire-extinguishing materials to control and limit fire spread.
- Boost the concept and culture of environmental sustainability of ecosystems.

Other sections of this paper are organized as follows: the next section presents background and literature review on forest fire detection, drones utilization, and new technologies for environmental sustainability. The third section presents the proposed system framework, including the system architecture, spiral coverage algorithm, UAV swarm operational mechanism, VLM analysis model, and early fire detection mechanism. The fourth section describes the proposed system and its application scenarios. The fifth section concludes the paper and presents future work.

II. RELATED WORK

In recent years, there has been growing interest in developing smart solutions for early detection and mitigation of forest fires, given climate change and increasing temperatures. These solutions differ from the ground-based and satellite monitoring systems that have been used for years to the more recent systems using AI and UAVs. In this section, we review the most recent research to date on these topics [19].

A. IoT-based Fire Detection Systems

Many studies have made use of Internet of Things (IoT) networks to monitor forests for temperature, humidity, and smoke levels. This data is sent to the central data analysis center for examination to see if there are abnormal variations that indicate the beginning of a fire. While they work in some situations, these systems have a number of challenges, such as high installation and maintenance costs, extensive infrastructure, and low coverage in dense natural forests or remote areas. They are also prone to failures or lost connection in harsh environments [20, 21].

B. Satellite-based Monitoring

Satellite-based monitoring is one of the primary tools for monitoring large-scale forest fires and provides periodic images of vast landmasses. These systems have helped to improve post-ignition fire tracking. However, they have limited image resolution, time-dependent updates, and failure to detect fires in early stages, especially in regions of heavy cloud cover or dense vegetation [22].

C. UAV-based Forest Monitoring

With advancements in UAV technologies, many studies have been conducted on their applications in environmental monitoring and the detection of fire. UAVs are very portable and can travel across difficult terrain, and they can retain high-resolution images in real time. Some work has been done on drone-based monitoring of a specific area. Some have been done on UAV swarms to enhance coverage and monitoring efficiency. Despite these advances, most of the solutions are based on monitoring and not in conjunction with any preventive effort, and do not provide proper coverage strategies. Furthermore, some systems lack smart coverage strategies that guarantee continuous and systematic surveillance of target areas [23].

D. Artificial Intelligence and Image Analysis for Fire Detection

The field of fire detection has received much help with the use of AI technologies using both images and video analysis. Deep learning models have been applied to detect smoke and

flames in aerial images or videos to improve detection accuracy and reduce response times. Vision-language models (VLMs) have recently been introduced and provide deeper, more accessible visual and linguistic understanding of complex scenes, thanks to pre-trained models. However, these studies are largely analytical, and they do not incorporate these models to build complete operational systems that depend on continuous tracking and quick decision-making [24-26].

E. Research Gap

While there has been huge progress in IoT, satellite monitoring, UAVs, and AI image analysis, integrated systems still have a serious gap to fill, including:

- Continuous and effective coverage of vast forested areas
- Real-time smart analysis using sophisticated models like VLMs.
- A fast preventive response mechanism to stop fire spreading.
- Cost-effective, scalable solutions that can be applied in natural environments.

In this paper, we address this issue by introducing an integrated platform that is based on a UAV swarm, smart coverage algorithm, deep visual analysis, and an early preventive response mechanism.

III. METHODOLOGY

We propose to develop a multi-layered framework for early detection and prevention of forest fires from UAVs, Fog Computing, Cloud Computing, and a smart application layer. This framework would enable monitoring, rapid analysis, and preventive response in the large-scale natural environment.

The proposed system consists of four main layers, as shown in Figure 1.

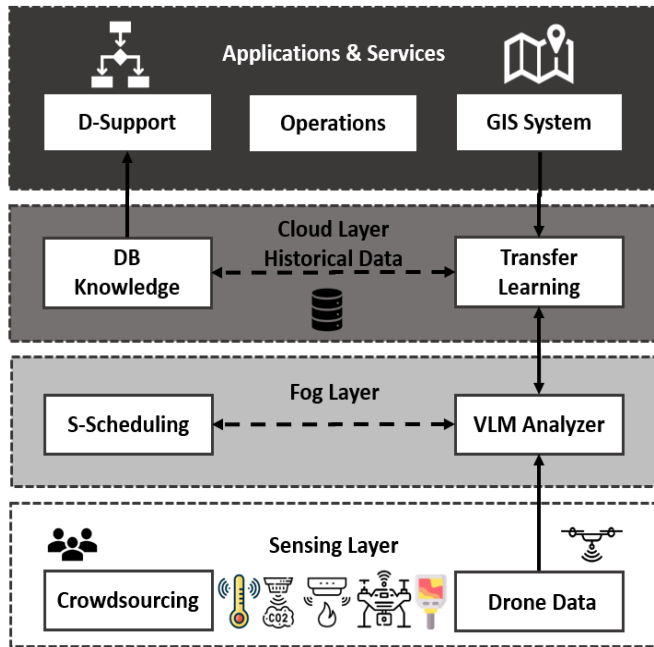


Fig. 1. Process Flow

A. Sensing Layer

This layer is the field interface of the system and relies on a swarm of UAVs equipped with mobile sensors (e.g., smoke, CO₂, and temperature sensors), as well as high-resolution cameras (RGB and thermal cameras) that monitor the environment. These UAVs continuously observe the environment, including aerial images, temperatures, and smoke or abnormal changes in vegetation cover.

It is also a very flexible layer because we can deploy the UAVs in large areas and hard to reach and dynamically change the coverage density with environmental conditions (e.g., high temperatures or higher fire risk). In addition, the swarm nature of the UAVs is good for balanced coverage and minimizes the time between monitoring.

B. Fog Layer

The fog computing layer is a key component of the solution and is based on a UAV ground station that serves as a Fog node. This layer provides data processing close to the data source, minimizing latency and enabling decisions in real time. The importance of fog nodes is that they reduce the time required to send data to the cloud, allow for real-time analysis of the images and data, provide quick response to emergency situations, and have low energy and bandwidth consumption.

This layer is based on the use of Application Programming Interfaces (APIs) and VLMs to analyze images captured by the UAVs and enhanced sensor readings, thus detecting early signs of fire such as smoke, flames, or abnormal environmental events. The fog nodes handle the swarm management by scheduling the UAVs for each of the coverage areas, the flight paths, and avoiding overlap in the coverage areas and redundant UAVs.

The proposed algorithm is based on spiral coverage to ensure comprehensive and systematic monitoring of forested areas. The area is organized into center points. Each UAV starts from the center point and then moves along a spiral path to gradually cover the entire area from the inside out.

1) Operational Mechanism

- Defining the center of the monitoring area for each UAV.
- The UAV is moving in a circular path with a small initial radius and gradually expands after each rotation.
- The spiral moves on until the entire zone is covered.
- The UAV moves to a new area, or the cycle can be restarted after recharging.
- Each time a new UAV is launched, make sure coverage is complete and continuous with a UAV swarm.

2) Algorithm Advantages

- Filling the gaps in coverage.
- Assessing the same areas regularly.
- Increasing energy efficiency.
- Eliminating interference between UAVs.

Further, the proposed framework adapts to the environment to reduce costs and improve efficiency by setting the coverage speed of the UAV and the number of active UAVs according to the temperature, wind speed, and the risk level in the region.

C. Cloud Layer

It is the main storage layer of data and has high computation power for analysis and processing. All the drones' data that are collected and received from fog nodes will be saved permanently here. The cloud layer enables the platform to create and build real historical databases, which will be utilized for training the ML models with continuous tuning (Model Tuning) of the VLMs. These algorithms will contribute to the study of fire patterns and behavior. Thus, it helps to make the system accurate and adaptive over time by tuning and adapting to different environmental conditions.

D. Application Layer

It is the end-user interface of the proposed framework. It provides many services to various parties, such as the forest protection governmental agencies, drone operating companies, and first aid or emergency teams. It also enables users and volunteers to share with the crowdsourcing services, and allows them to report potential fires to the system via the smart mobile application. This data is integrated with other IoT and drone data for boosting detection accuracy and rapid response.

The following pseudocode gives a brief glimpse of the spiral rotation algorithm for a UAV:

Input: $a, b, \theta_{\max}, \Delta\theta$ // where a is the initial radius, and b is the rate of expansion, r is the new radius, while θ represents the angle of rotation about the center of the observation area.

Output: Sequence of UAV positions (x, y)

```

1: Initialize  $\theta = 0$ 
2: while  $\theta \leq \theta_{\max}$  do
3:    $r = a + b\theta$ 
4:    $x = r \cos(\theta)$ 
5:    $y = r \sin(\theta)$ 
6:   Move UAV to position  $(x, y)$ 
7:   Capture image/data at current position
8:    $\theta = \theta + \Delta\theta$ 
9: end while

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In summary, the proposed framework combines smart field monitoring using UAVs, real-time analysis of the sensor using fog computing, and long-term storage and learning in the cloud with application services. This integration allows for the development of an effective and scalable system for early detection and control of forest fires and environmental sustainability at the same time, and thus a sustainable system. Figure 2 presents the proposed drone movement path as proposed by the methodology.

IV. RESULTS AND DISCUSSION

At this stage, we were working on a framework and testing its utility and not doing a full system simulation. The aim of this work is to provide a detailed picture of the proposed

system in terms of architecture, algorithms, and mechanisms, and a preliminary assessment of one of its key components: image analysis using AI technologies.

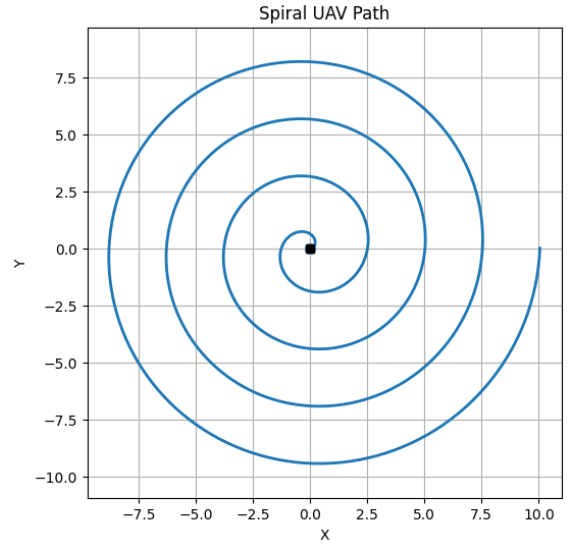


Fig. 2. Proposed UAV Movement Path

The UAVs capture aerial images and analyze them using AI models, which are able to understand the data through semantic scene understanding instead of just seeing the visual patterns. Danger indicators are identified, including light smoke emerging from the sky and not visible to the naked eye, discoloration of the vegetation, abnormal thermal hotspots, the initial formation of small flames, or the presence of fire in the area of a fire.

This allows early detection of the incident before the full ignition process has started. Once a hazard is detected, the location is identified, and a fast-response UAV is sent to spray fire-retardant foam over the hotspot to prevent the spread of the fire.

We have performed a first experiment using an API for an advanced Vision-Language Model based on ChatGPT architecture (GPT-5.3) to analyze aerial images and find fire indicators. We tested the model on a specialized dataset, Fire-Detection-Image-Dataset-master [27].

This dataset contains a variety of images, including actual fire scenes and misleading images that show either colors or patterns that look like fire, such as sunsets or red twilight skies (refer Figure 3).

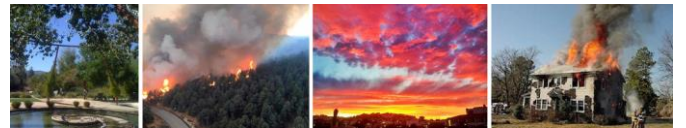


Fig. 3. Sample Images from a specialized Fire Detection Dataset

The model is able to describe the data with high accuracy and achieved a fire detection accuracy of over 99% in binary classification (presence or absence of fire). Also, we were able to distinguish between real fires versus natural scenes that might cause visual confusion (sunset imagery or red skies), and

this showed the power of the Vision-Language Model in this context. Figure 4 shows a conceptual visualization of the model in a real-world situation.

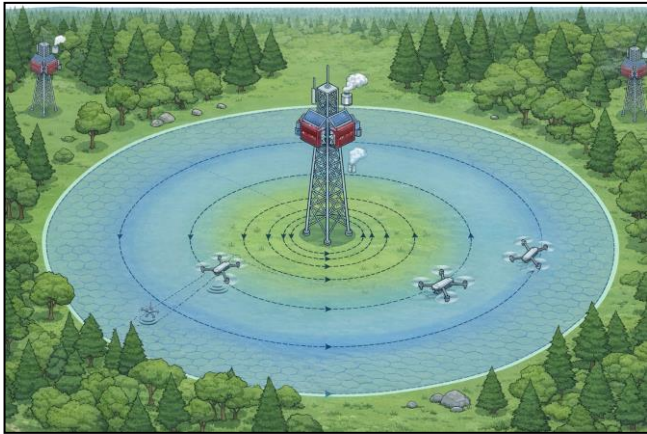


Fig. 4. Conceptual Model of the Coverage Process

However, like any other work and any use of modern technologies, there are still challenges that need to be solved in the future. The most important challenges are security [28-32], privacy [33-44], and interoperability [45]. And the scheduling and auto-selecting algorithms [46, 47] still have their scope for researchers, especially with the adoption and integration of fog computing with cloud [48]. In addition, our methodology can be used in different applications like monitoring crowds [49, 50] or disaster cases [51, 52].

V. CONCLUSION

We have proposed a novel framework combining UAV swarms and AI technologies to track forest fires ultra-early with a spiral coverage algorithm to ensure systematic and effective monitoring. Vision-language models (VLMs) were employed to analyze aerial imagery and detect early signs of fire, and hence to make early detection possible to control fire spreading at early stages. The framework shifts the reactive models to proactive ones, where it uses prediction and early intervention. This proactive approach can protect the forest from fires and ensure more sustainability. In future work, we propose to add an additional layer (semantic analysis layer) based on LLM-Assisted. This layer will enable the framework to filter based on a semantic classification to improve accuracy and reduce false alerts. Moreover, we will apply a proof-of-concept (PoC) for testing in a real-world environment to evaluate the system. Finally, it is important to develop a smartphone application that enables volunteers to report suspicious cases with auto-classifying reports. All reports and actions will be presented in an interactive dashboard for decision makers in real-time to assist decision-making efficiently.

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